UNIT 7

WAVE OPTICS

An age is called dark, not because the light fails to shine, but because people refuse to see.
— James Albert Michener

Learning Objectives

In this unit, the students are exposed to,

- The wave aspect of light.
- The proof for law of reflection and refraction.
- The phenomena like interference, diffraction and polarisation.
- The terms like magnification and resolving power etc.
- The different optical instruments like microscope, telescope etc.

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7.1 THEORIES ON LIGHT

Light is a form of energy that is transferred from one place to another. A glance at the evolution of various theories of light put forth by scientists will give not only an over view of the nature of light but also its propagation and some phenomenon demonstrated by it.

7.1.1 Corpuscular theory

Sir Isaac Newton (1672) gave the corpuscular theory of light which was also suggested earlier by Descartes (1637) to explain the laws of reflection and refraction. According this theory, light is emitted as tiny, massless (negligibly small mass) and perfectly elastic particles called corpuscles. As the corpuscles are very small, the source of light does not suffer appreciable loss of mass even if it emits light for a long time. On account of high speed, they are unaffected by the force of gravity and their path is a straight line in a medium of uniform refractive index. The energy of light is the kinetic energy of these corpuscles. When these corpuscles impinge on the retina of the eye, the vision is produced. The different sizes of the corpuscles give different colours to light. When the corpuscles approach a surface between two media, they are either repelled (or) attracted. The reflection of light is due to the repulsion of the corpuscles by the medium and refraction of light is due to the attraction of the corpuscles by the medium.

This theory could not explain the reason why the speed of light is lesser in denser medium than in rarer medium and also the phenomena like interference, diffraction and polarisation.

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